

Complex Analysis

Home Work 1

Student: Mark Roberts [26-712-224]

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Question 1.1

- (a) Find the modulus and the principal branch of argument of the following complex numbers. Hence write them in polar form.

- (i) $1 + i$
- (ii) $6 + 2\sqrt{3}i$
- (iii) $\frac{\sqrt{2}}{1-i}$

- (b) Write the following complex numbers in the form $a + bi$:

- (i) $2e^{3\pi i/4}$
- (ii) $e^{7\pi i/6}$
- (iii) $\frac{1+i}{i} + \frac{i}{1-i}$

- (c) Write the following complex numbers in the form $a + bi$, and in polar form:

- (i) $\frac{1+i}{\sqrt{3}-3i}$
- (ii) $(1-i)^3$
- (iii) $\left(\sqrt{2+\sqrt{2}} + \sqrt{2-\sqrt{2}}i\right)^2$

Using your answer to (iii), deduce the value of $\cos \frac{\pi}{8}$.

ANSWER:

- (a.i) $|1+i| = \sqrt{1^2+1^2} = \sqrt{2}$
- (a.ii) $|6+2\sqrt{3}i| = \sqrt{6^2+(2\sqrt{3})^2} = \sqrt{36+12} = 4\sqrt{3}$
- (a.iii) $\left|\frac{\sqrt{2}}{1-i}\right| = \frac{\sqrt{2}}{|1-i|} = \frac{\sqrt{2}}{\sqrt{1^2+(-1)^2}} = 1$
- (b.i) $2e^{3\pi i/4} = 2(\cos(3\pi/4) + i\sin(3\pi/4)) = 2\left(\frac{1}{\sqrt{2}}\right)(-1+i) = -\sqrt{2} + \sqrt{2}i$
- (b.ii) $e^{7\pi i/6} = (\cos(7\pi/6) + i\sin(7\pi/6)) = \left(-\frac{\sqrt{3}}{2} - \frac{1}{2}i\right)$
- (b.iii)

$$\begin{aligned}\frac{1+i}{i} + \frac{i}{1-i} &= \left(\frac{-i}{-i}\right) \frac{1+i}{i} + \left(\frac{1+i}{1+i}\right) \frac{i}{1-i} \\ &= (1-i) + \frac{-1+i}{2} \\ &= \frac{1-i}{2}\end{aligned}$$

(c.i)

$$\begin{aligned}\frac{1+i}{\sqrt{3}-3i} &= \left(\frac{\sqrt{3}+3i}{\sqrt{3}+3i} \right) \left(\frac{1+i}{\sqrt{3}-3i} \right) \\ &= \frac{(\sqrt{3}+3i)(1+i)}{(\sqrt{3})^2+3^2} \\ &= \frac{(\sqrt{3}-3)+i(\sqrt{3}+3)}{12} \\ &= \frac{\sqrt{3}-3}{12} + i \frac{\sqrt{3}+3}{12}\end{aligned}$$

(c.ii)

$$\begin{aligned}(1-i)^3 &= (1-i)^2(1-i) \\ &= -2i(1-i) \\ &= -2-2i\end{aligned}$$

(c.iii)

$$\begin{aligned}\left[\sqrt{2+\sqrt{2}} + \sqrt{2-\sqrt{2}}i \right]^2 &= (2+\sqrt{2}) + 2 \left(\sqrt{2+\sqrt{2}} \right) \left(\sqrt{2-\sqrt{2}} \right) i - (2-\sqrt{2}) \\ &= 2\sqrt{2} + 2\sqrt{4-2}i \\ &= 2\sqrt{2} + 2\sqrt{2}i\end{aligned}$$

Question 1.2

(a) What do the following subsets represent geometrically? Draw a sketch of each one in the complex plane.

(i) $|z + 1 + i| = 1$

(ii) $1 < \operatorname{Re} z \leq 2$

(iii) $|z - 2| < 3$ and $|\operatorname{Im} z| > 1$

(iv) $\operatorname{Im} z^2 \geq 0$

(b) Which of the above sets are

(i) Open?

(ii) Closed?

ANSWER:

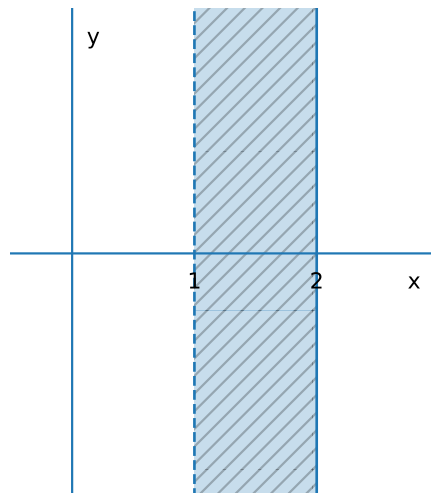
(a.i) $|z + 1 + i| = 1 \Rightarrow |z - (-1 - i)| = 1$ This represents a circle with centre $(-1 - i)$ and radius 1.

(a.ii) $1 < \operatorname{Re}(z) \leq 2$ represents a vertical strip in the Argand diagram.

The region in question is shown shaded.

$x = 1$ **is not** in the region.

$x = 2$ **is** in the region.

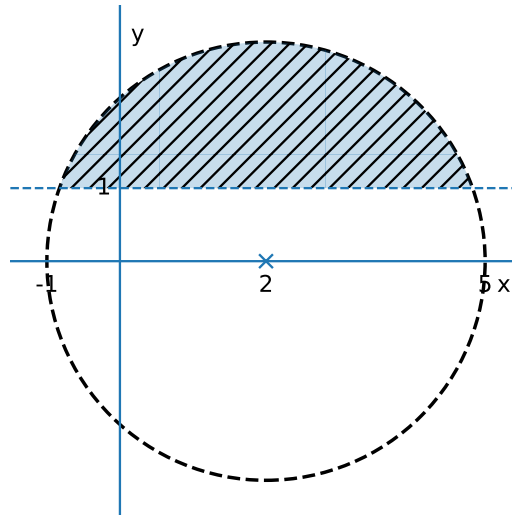


(a.iii) $|z - 2| < 3$ represents the inside of a circle centred at $2 + 0i$ with radius 3.

$|\operatorname{Im}(z)| > 1$ represents the section of the Argand diagram for which $y > 1$.

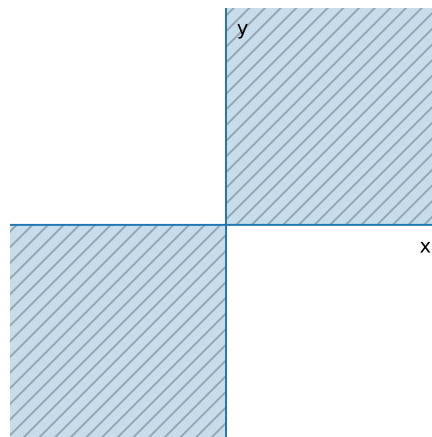
Any $z \in \mathbb{C}$ that satisfies both conditions will be in the inside of the circle (centred on $2 + 0i$, with radius 3) and lies above the line $y = 1$.

The shaded region in the diagram is the required region.



(a.iv) $\operatorname{Im} z^2 \geq 0$. Let $z = x + yi \Rightarrow z^2 = x^2 - y^2 + 2xyi \Rightarrow \operatorname{Im}(z^2) = 2xy$.

Now $2xy \geq 0$ is given by the 2 quadrants shown below.



(b)(i) The boundary of a circle is a **closed** set.

(ii) The boundary is open on one side ($x = 1$) & closed on the other ($x = 2$). So the region is **neither open nor closed**.

(iii) The two sets in question are both open. This $\Rightarrow$ their intersection will also be an **open** set.

(iv) The $x = 0$ & $y = 0$ are included in the region (since the two regions contain all their limit pts) $\Rightarrow$ region is a **closed** set.

Question 1.3 Find all roots in $\mathbb{C}$ of the following equations (you may express your answer in either $a + bi$ or polar form, whichever you prefer):

(i) $z^3 + 1 = 0$

(ii) $1 + z + z^2 = 0$ *Hint: First try expanding $(z - 1)(1 + z + z^2)$.*

(iii) $1 - z^2 + z^4 = 0$

ANSWER:

(i) $z^3 + 1 = 0$. Let $z = re^{i\theta}$. Then we get:

$$(re^{i\theta})^3 + 1 = 0$$

$$\Rightarrow r^3 e^{3i\theta} = -1 \tag{A}$$

As $r \geq 0 \Rightarrow r = 1$. Using (A) we can determine the θ values as:

$$\theta_n = \frac{\pi(2n+1)}{3}, \quad n \in \{0, 1, 2\}$$

From which:

$$\theta_0 = \frac{\pi}{3}, \quad \theta_1 = \pi, \quad \theta_2 = \frac{5\pi}{3}$$

We can now calculate the z_n :

$$z_0 = e^{i\pi/3} = \frac{1}{2} + \frac{\sqrt{3}}{2}i$$

$$z_1 = e^{i\pi} = -1$$

$$z_2 = e^{i5\pi/3} = \frac{1}{2} - \frac{\sqrt{3}}{2}i$$

(ii) Clearly $1 + z + z^2 = 0$ is not satisfied by $z = 1$.

Now:

$$(z - 1)(1 + z + z^2) = z^3 - 1 \tag{B}$$

Now the solutions of (B) are just the solutions of $1 + z + z^2 = 0$ and $z = 1$ and follow the same sort of pattern as (i) above, but in this case: $z^3 = 1 = e^{i0}$.

These solutions are:

$$\theta_n = \frac{2n\pi + 0}{3}, \quad n \in \{0, 1, 2\}$$

We can ignore $\theta_0 = 0$, as this is the $z = 1$ solution. The solutions to $1 + z + z^2 = 0$ are found when $n = 1, 2$:

$$\begin{aligned} z_1 &= e^{i2\pi/3} = -\frac{1}{2} + \frac{\sqrt{3}}{2}i \\ z_2 &= e^{i4\pi/3} = -\frac{1}{2} - \frac{\sqrt{3}}{2}i \end{aligned}$$

(iii) $1 - z^2 + z^4 = 0$ can be simplified by replacing $y = -z^2$ in which case we would then have the equivalent equation:

$$1 + y + y^2 = 0 \tag{C}$$

We have already solved this equation in (ii) above. The solutions being: $y_1 = -z_1^2 = e^{i2\pi/3}$ and $y_2 = -z_2^2 = e^{i4\pi/3}$. Or, equivalently: $z_1^2 = e^{i5\pi/3}$ and $z_2^2 = e^{i7\pi/3}$.

Each solution has two separate solutions for z - so we have a total of 4 solution:

$$\begin{aligned} z_{11} &= e^{i5\pi/6} &= -\frac{\sqrt{3}}{2} + \frac{1}{2}i \\ z_{12} &= -e^{i5\pi/6} &= \frac{\sqrt{3}}{2} - \frac{1}{2}i \\ z_{21} &= e^{i7\pi/6} &= -\frac{\sqrt{3}}{2} - \frac{1}{2}i \\ z_{22} &= -e^{i7\pi/6} &= +\frac{\sqrt{3}}{2} + \frac{1}{2}i \end{aligned}$$

Question 1.4 For each of the following, either find the limit or prove that it does not exist, as appropriate.

(i) $\lim_{z \rightarrow 3+4i} (z - 7i)^2$

(ii) $\lim_{z \rightarrow 0} \frac{\operatorname{Re} z + \operatorname{Im} z}{z}$

(iii) $\lim_{z \rightarrow 2i} \frac{z^2 + 4}{z - 2i}$

ANSWER:

(i)

As $(z - 7i)^2$ is a continuous function on $\mathbb{C}$, we can replace z by $3 + 4i$:

$$\lim_{z \rightarrow 3+4i} (z - 7i)^2 = (3 + 4i - 7i)^2 = (3 - 3i)^2 = -18i$$

(ii)

$$\frac{\operatorname{Re}(z) + \operatorname{Im}(z)}{z} = \frac{x + y}{x + iy}$$

Hence:

$$\lim_{z \rightarrow 0} \frac{\operatorname{Re}(z) + \operatorname{Im}(z)}{z} = \lim_{z \rightarrow 0} \frac{x + y}{x + iy}$$

But this has no limit, since along the real line the limit is 1 and along the imaginary line the limit is i .

(iii)

$$z^2 + 4 = (z + 2i)(z - 2i)$$

$$\Rightarrow \lim_{z \rightarrow 2i} \frac{z^2 + 4}{z - 2i} = \lim_{z \rightarrow 2i} (z + 2i) = 4i$$

Question 1.5 For each of the following functions, write the real and imaginary parts as functions of (x, y) , where $z = x + iy$.

(i) $z^3 + 2z + 1$

(ii) $z + z^{-1}$ (for $z \neq 0$)

(iii) $\frac{1}{1-z}$ (for $z \neq 1$)

ANSWER:

(i)

$$\begin{aligned} z^3 &= (x + iy)^3 = (x + iy)(x^2 - y^2 + 2xyi) \\ &= x(x^2 - y^2) - 2xy^2 + iy(x^2 - y^2) + 2x^2yi \\ &= x(x^2 - 3y^2) + iy(3x^2 - y^2) \end{aligned}$$

$$\Rightarrow z^3 + 2z + 1 = [x(x^2 - 3y^2 + 2) + 1] + i[y(3x^2 - y^2 + 2)]$$

From which we can determine the real and imaginary functions as:

$$\begin{aligned} u(x, y) &= x(x^2 - 3y^2 + 2) + 1 \\ v(x, y) &= y(3x^2 - y^2 + 2) \end{aligned}$$

Where: $u(x, y)$ is the real function and $v(x, y)$ is the imaginary function.

(ii) For $z \neq 0$ we have:

$$\begin{aligned} z + \frac{1}{z} &= (x + iy) + \frac{1}{x + iy} \\ &= (x + iy) + \frac{x - iy}{x - iy} \frac{1}{x + iy} \\ &= (x + iy) + \frac{x - iy}{x^2 + y^2} \\ &= \left(x + \frac{x}{x^2 + y^2}\right) + i\left(y - \frac{y}{x^2 + y^2}\right) \\ &= x\left(1 + \frac{1}{x^2 + y^2}\right) + iy\left(1 - \frac{1}{x^2 + y^2}\right) \end{aligned}$$

From which we determine that:

$$u(x, y) = x\left(1 + \frac{1}{x^2 + y^2}\right) \quad v(x, y) = y\left(1 - \frac{1}{x^2 + y^2}\right)$$

are the real & complex functions respectively.

(iii) For $z \neq 1$ we have:

$$\begin{aligned}\frac{1}{1-z} &= \frac{1}{(1-x)-iy} \\ &= \frac{1}{(1-x)-iy} \frac{(1-x)+iy}{(1-x)+iy} \\ &= \frac{(1-x)+iy}{(1-x)^2+y^2}\end{aligned}$$

From which we determine that:

$$u(x, y) = \frac{(1-x)}{(1-x)^2+y^2} \quad v(x, y) = \frac{y}{(1-x)^2+y^2}$$

Question 1.6 Prove that, for all $z, w \in \mathbb{C}$,

$$|z + iw|^2 + |w + iz|^2 = 2(|z|^2 + |w|^2).$$

Extra: Can you see a geometric interpretation of this identity?*

ANSWER:

$$|z + iw|^2 = (z + iw)(\bar{z} - i\bar{w})$$

$$= |z|^2 + |w|^2 + iw\bar{z} - iz\bar{w}$$

$$|w + iz|^2 = (w + iz)(\bar{w} - i\bar{z})$$

$$= |w|^2 + |z|^2 + iz\bar{w} - iw\bar{z}$$

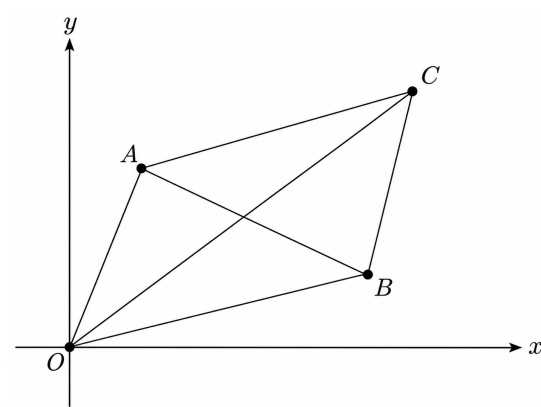
Note that the terms in the expansion that contain i have opposite signs - so summing the two terms will cause them to cancel:

$$\Rightarrow |z + iw|^2 + |w + iz|^2 = 2|z|^2 + 2|w|^2$$

$$= 2(|z|^2 + |w|^2)$$

□

This formula is nothing more than the parallelogram law.



This law states that for a parallelogram (shown above) we have:

$$2(|OA|^2 + |OB|^2) = |AB|^2 + |OC|^2 \quad (1.6A)$$

Take $z = OB$ and $iw = OA$ and noting that $|iw| = |w|$. Then (1.6A) becomes:

$$2(|z|^2 + |w|^2) = |z + iw|^2 + |z - iw|^2 \quad (1.6B)$$

This is already near the formula we want [i.e. $|z + iw|^2 + |w + iz|^2 = 2(|z|^2 + |w|^2)$]; we only need to show that $|z - iw| = |w + iz|$. Now:

$$\begin{aligned} w + iz &= i(z - iw) \\ \implies |w + iz| &= |i(z - iw)| \\ &= |z - iw| \end{aligned}$$

□

Question 1.7 Show that, for all $z, w \in \mathbb{C}$ such that $|z| < 1$ and $|w| < 1$,

$$\left| \frac{z - w}{1 - \bar{z}w} \right| < 1.$$

Now assume that $z, w \in \mathbb{C}$ with $1 - \bar{z}w \neq 0$ (but not necessarily that $|z| < 1$ and $|w| < 1$). For which z, w do we have

$$\left| \frac{z - w}{1 - \bar{z}w} \right| = 1?$$

ANSWER:

$z, w \in \mathbb{C}$ and $|z| < 1$ & $|w| < 1$. Now:

$$\left| \frac{z - w}{1 - \bar{z}w} \right| < 1$$

$$\Leftrightarrow |z - w| < |1 - \bar{z}w|$$

$$\Leftrightarrow |z - w|^2 < |1 - \bar{z}w|^2$$

$$\Leftrightarrow (z - w)(\bar{z} - \bar{w}) < (1 - \bar{z}w)(1 - z\bar{w})$$

$$\Leftrightarrow |z|^2 - z\bar{w} - \bar{z}w + |w|^2 < 1 - z\bar{w} - \bar{z}w + |z|^2|w|^2$$

$$\Leftrightarrow 0 < 1 - |z|^2 - |w|^2 + |z|^2|w|^2$$

$$\Leftrightarrow 0 < (1 - |w|^2)(1 - |z|^2) \quad (***)$$

As $|z| < 1$ & $|w| < 1$

$$(1 - |w|^2) > 0 \quad \& \quad (1 - |z|^2) > 0.$$

And hence equation (***) holds. Consequently:

$$\Rightarrow \left| \frac{z - w}{1 - \bar{z}w} \right| < 1.$$

For equality to hold

$$\left| \frac{z-w}{1-\bar{z}w} \right| = 1 \Leftrightarrow |z-w|^2 = |1-\bar{z}w|^2$$

$$\Leftrightarrow |w|^2 + |z|^2 = 1 + |z|^2|w|^2$$

$$\Leftrightarrow 0 = 1 - |w|^2 - |z|^2 + |z|^2|w|^2$$

$$= (1 - |w|^2)(1 - |z|^2)$$

$$\Rightarrow |w| = 1 \quad \text{or} \quad |z| = 1.$$

We need to qualify this last result as we must also have that $1 - \bar{z}w \neq 0$. So we must exclude the condition $z = w$.

Question 1.8 Prove Cantor's Intersection Theorem (Theorem 1.44 in the Lecture Notes).

Theorem 1.44 (Cantor's Intersection Theorem). *Let $K \subset \mathbb{C}$ be a compact set, and let $\{C_n\}_{n=1}^{\infty}$ be non-empty closed sets with $C_n \subset K$ for all $n \in \mathbb{N}$, that are nested such that $C_{n+1} \subset C_n$ for all $n \in \mathbb{N}$. Then*

$$\bigcap_{n=1}^{\infty} C_n \neq \emptyset.$$

ANSWER:

From the definition of $\{C_n\}$ in the theorem it is clear that

$$C_n = \bigcap_{i=1}^n C_i \quad (\neq \emptyset) \quad \forall n \in \mathbb{N}$$

As K is compact we know that it is sequentially compact.

Construct a Sequence $\{z_n\}$ by selecting each z_n from C_n ($n = 1, 2, \dots$). Then, as $\{z_n\}$ is a sequence in the sequentially compact set K :

$$\Rightarrow \exists \{z_{n_k}\} \subset \{z_n\} \text{ that converges to } z \in K$$

But $\{z_{n_k}\}$ is also a convergent subsequence in each C_n and, as C_n is closed, $z \in C_n$. This is true for every C_n , $n \in \mathbb{N}$. Hence

$$z \in \bigcap_{i=1}^{\infty} C_n \quad \Rightarrow \quad \bigcap_{i=1}^{\infty} C_n \neq \emptyset$$

□